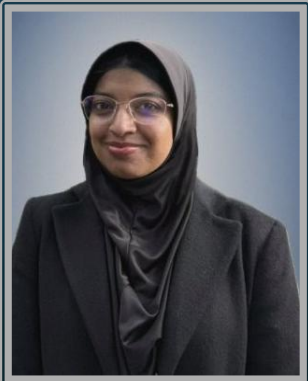


Multi-User Physiological Computing



Dr. Chris Crawford



Zara Ahmed



Trenton Ottman



Jackson Wallace



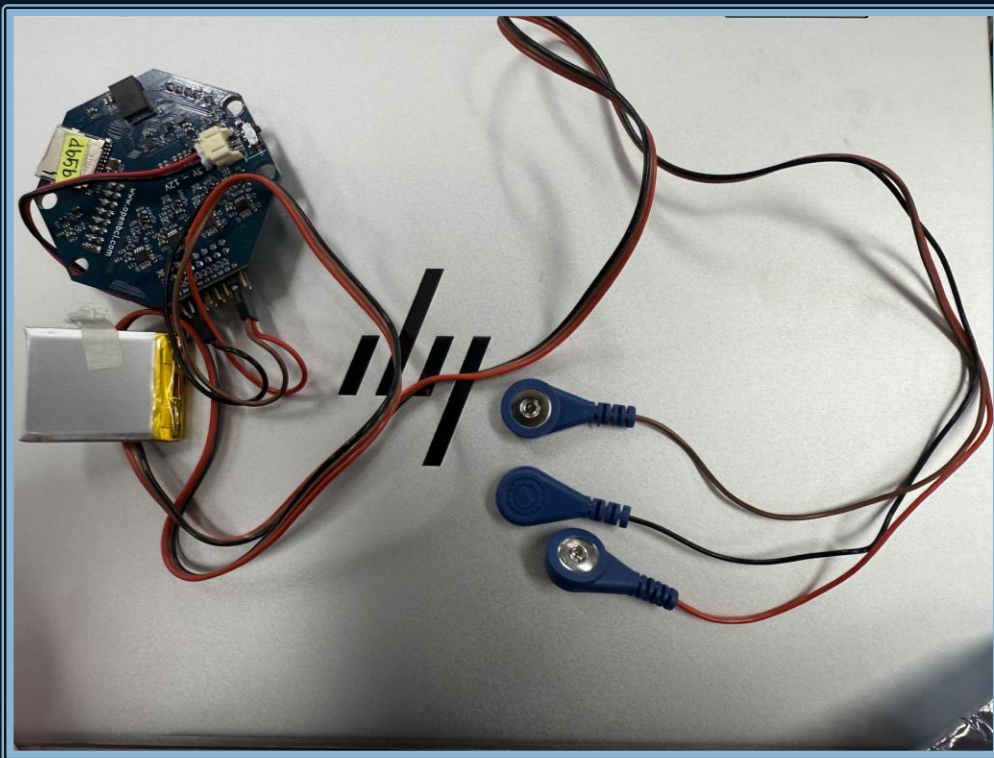
Ansel Stinnett



James Adler

Speaker:
Everyone

Project Review



- Continuation of previous CS 495 project “PhysioGaming”.
- Goal: Create a multiplayer game utilizing EMG sensors.
- Iteration 1: Proof of concept multiplayer implementation.

Speaker:
James Adler

Planning & Actualizing

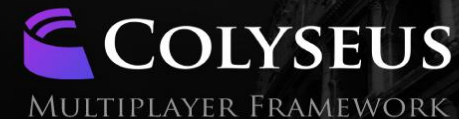
- Highest priority: working multiplayer
- Research game ideas, multiplayer options and game engines
- Create an MVP of a playable multiplayer game

Iteration 1 Planning

Iteration Goal

- Multiplayer prototype/proof-of-concept
- Rudimentary game design consideration

- Decided to use Colyseus for multiplayer
- Decided not to use Unity
- Created MVP of pong for the demo

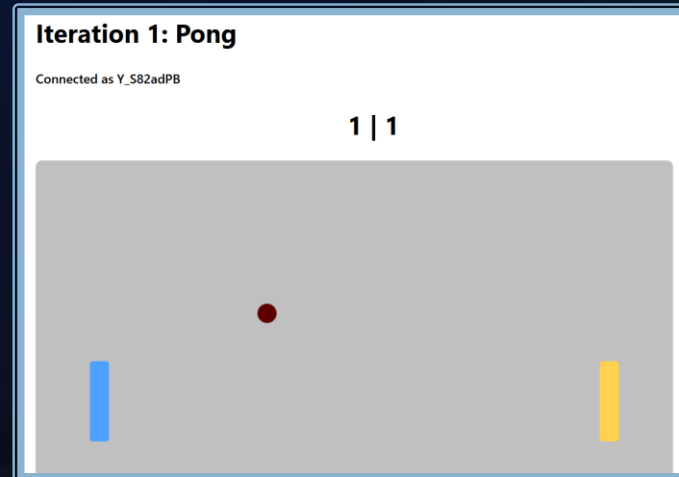


Speaker:
Jackson Wallace

Major Accomplishments

- Established team development practices
- Initialized a Colyseus client/server architecture
- Refactored the inherited repository to support our project's requirements
- Built a working multiplayer prototype
- Deployed the multiplayer server to Google Cloud

Steps	Duration
✓ Build Summary 4 Steps	00:02:59
✓ 0: Buildpack pack build us-central1-docker.pkg.dev/physi...	00:01:27
✓ 1: Pull pull us-central1-docker.pkg.dev/physiogami...	00:00:18
✓ 2: Push push us-central1-docker.pkg.dev/physiogam...	00:00:02
✓ 3: Deploy gcloud run services update cs495-multiuser...	00:01:02



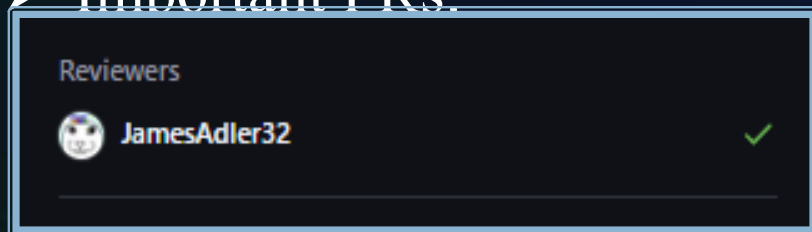
Speaker:
Trenton Ottman

Engineering & Quality

➤ Investigations



➤ Important PRs:



on

docs

multiplayer-implementation

- feat: removed playground and monitor links**
#38 · LocandLoad opened 19 hours ago · Closed 8 hours ago Collaborator · Approved
- PBI-05 Iteration 1 demo**
#36 · JamesAdler32 opened yesterday · Closed yesterday Collaborator · Approved
- Setup Colyseus Server Config For Cloud Deployment**
#35 · TrentonOttman opened 2 days ago · Closed 2 days ago Owner · Approved
- Zaahmed0786 Docs & Unity Deletion**
#34 · zaahmed0786 opened 2 days ago · Closed 23 minutes ago Collaborator · Approved
- Setup Colyseus Server Config For Cloud Deployment**
#32 · TrentonOttman opened 2 days ago · Closed 2 days ago Owner · Approved

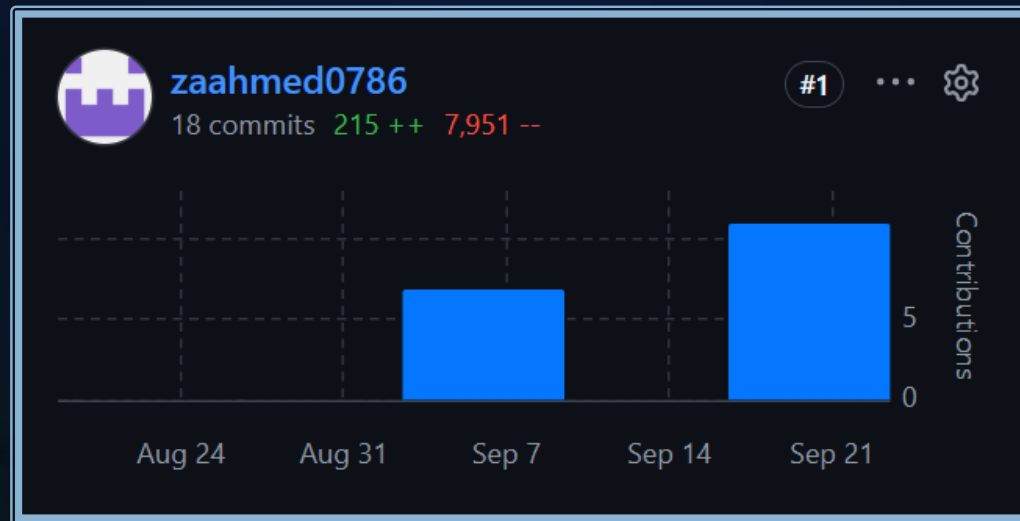
playroom.md

unity-multiplayer-services.md

Speaker:
Zara Ahmed

Team Contributions - Zara

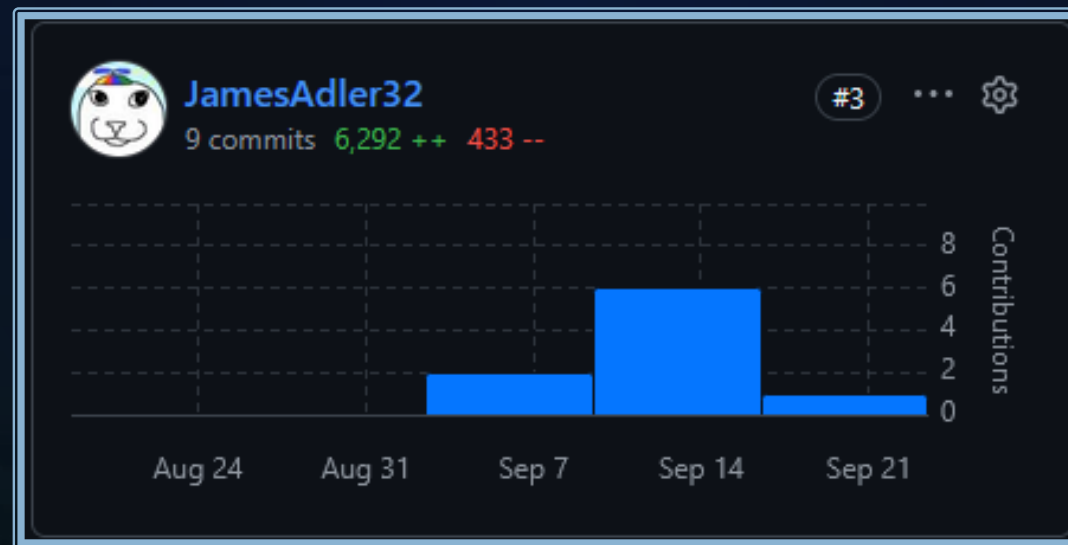
- Investigations for Glitch, Playroom, Unity, Godot, and 5 Game Design Ideas
- Minor Unity training and eventual removal
- Misc. documentation



Speaker:
Zara Ahmed

Team Contributions - James

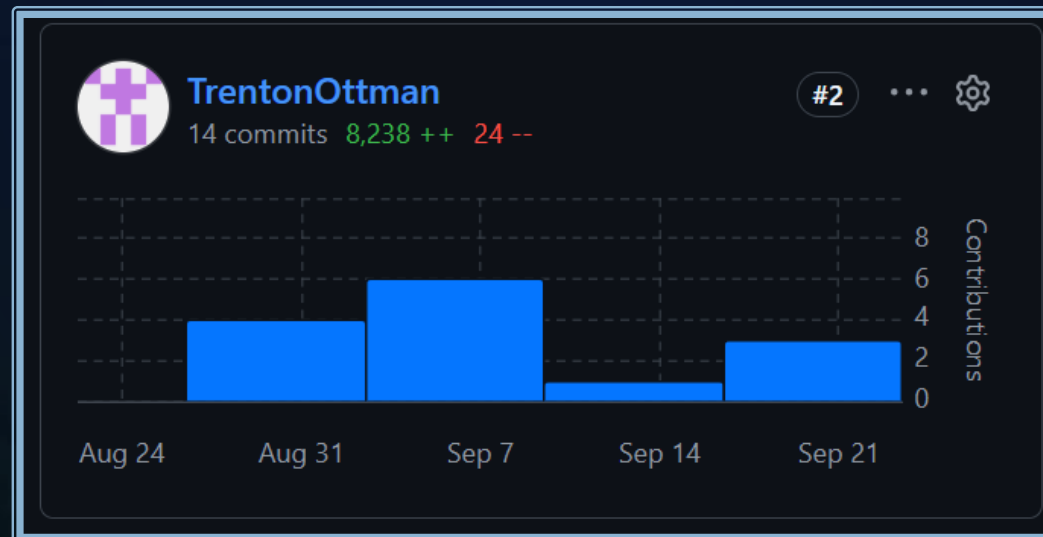
- Implemented the iteration 1 demo game
- Investigations for Colyseus, Unity, and other potential multiplayer tech stacks
- Misc. documentation



Speaker:
James Adler

Team Contributions - Trenton

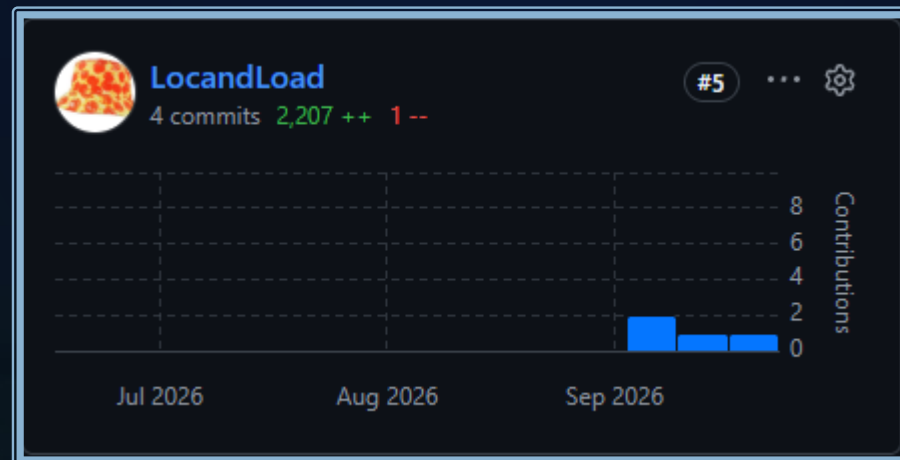
- Investigations for Colyseus, Unity, and Google Cloud.
- Set up Colyseus server configuration to work with a cloud deployment.
- Deployed Colyseus server to the cloud.



Speaker:
Trenton Ottman

Team Contributions - Jackson

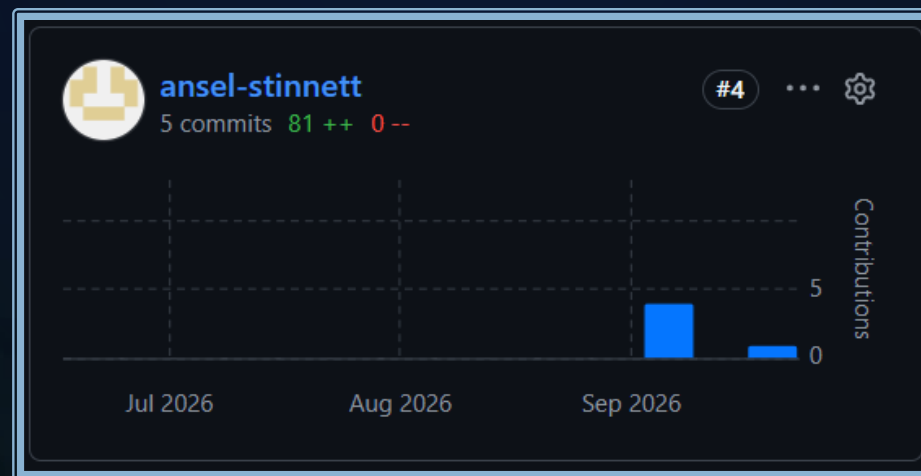
- Investigation for authoritative server architecture
- Archived and sifted through previous project files
- Made progress towards using EMG sensor for input



Speaker:
Jackson Wallace

Team Contributions – Ansel

- Reviewed previous team project to identify what we could reuse
- Helped brainstorm game design ideas
- Investigations for multiplayer approaches & scaling towards 8-12 players



Speaker:
Ansel Stinnett

Lessons Learned & Upcoming

- Colyseus handles real time data sync and room support
- Unity was disapproved of due to steep learning curve and sponsor's existing familiarity
 - Rather use TypeScript frameworks
- **Built a simple pong multiplayer game to prove that multiplayer works before adding complexity later.**
- **Iteration 2:** Integrate EMG sensors into next game iteration and scale a game toward 8 – 12 players

AI Usage

- Investigations
- Documentation
- Code assists

Stand-Up Update — Trenton

Previous commitment: Investigate cloud options

Outcome: Investigated cloud options

Evidence: docs/investigations/google_cloud.md

Since the last stand-up: Investigated Google Cloud, fixed a ruleset issue for PRs

Next commitment: Work on slides

Blockers / dependencies: N/A

Collaboration / help: N/A

Material AI use: AI helped with investigation for Google Cloud

Speaker:
James Adler



Questions?